

An open-source multicanonical sampler for 2D causal set quantum gravity: validation and a random-background lattice-gas comparison

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Abstract

We describe an open-source Wang-Landau/multicanonical (WL/MUCA) sampler for two-dimensional causal set quantum gravity, built to replace a beta-by-beta parallel-tempering (PT) scan whose located critical couplings turned out to be mostly right-censored and, in one tested case, not reproducible under a blind re-centering of the search grid. The sampler is validated against the published finite-size-scaling formula for the order-based smeared Benincasa-Dowker action at two system sizes and two couplings, and against an independent causal-set model -- Cunningham & Surya's dimensionally-restricted lattice gas -- including that model's own qualitative manifold-like/layered phase structure. We then build a random-background variant of the lattice gas (a quenched Poisson sprinkling of background sites in place of C&S's regular lattice) to address an open question C&S raise but do not pursue: whether their regular background biases the result. At $n=30$, five independent background realizations locate a critical coupling about 12% below a matched-pipeline, 5-chain-seed regular- lattice ensemble ($z=51.8$ -- not a single-seed artifact on either side), while the qualitative phase structure and the hot phase's detailed statistics are close to background-independent. A direct mechanism check rules out the simplest explanation (the random background reaching a deeper cold-phase action floor); the real mechanism is left as an open question. At $n=50$, the gap survives ($z=13.7$) but shrinks to about 7%, driven mainly by the absolute gap itself falling about 3x ($1.372 \rightarrow 0.460$), and the cold-phase observable difference most visible at $n=30$ nearly closes. The relative gap times n is nearly constant across the two sizes ($12.4\% \times 30 = 3.71$ vs. $7.2\% \times 50 = 3.58$), consistent with -- but, from two points, not proof of -- a finite-size effect falling off roughly as $1/n$; taken at face value this predicts about a 5.6% relative gap at $n=64$, the next size reachable without a bitset-engine rewrite. Getting a trustworthy $n=50$ result required finding and fixing two further sampler bugs invisible at $n=30$: an adaptive-histogram-widening step that filled new bins with a flat value instead of a gradient (creating an unphysical "cliff" the walker could get trapped behind), and -- after fixing that -- a genuine kinetically-isolated bin that no amount of reweighting could repair, fixed by switching to a fixed, pilot-quantile-based sampling window with no adaptive widening at all. We report several methodological pitfalls found and fixed along the way -- a `beta_c` definition mismatch, noise mistaken for a first-order double peak (twice, in two different ways), a units bug in the random-background model's causal relation, and the two Wang-Landau histogram-widening bugs above -- as cautionary detail for others

building similar samplers. Limitations include small system sizes ($N, n \leq 60$), a first-order barrier that is not cleanly resolved at any size tested, and only two system sizes checked for the background-dependence finding.

1. Introduction

A causal set is a locally finite partially ordered set, proposed as a discrete substrate for Lorentzian quantum gravity: the partial order encodes the causal structure, and continuum geometry is meant to emerge statistically from a sum (or path integral) over causal sets weighted by a discrete analogue of the Einstein-Hilbert action. The leading discrete action for this purpose is the Benincasa-Dowker (BD) action, built from counts of small order-intervals ("abundances") in the causal set, which converges to the continuum scalar-curvature action when evaluated on a Poisson sprinkling into a Lorentzian manifold [Benincasa & Dowker 2010, arXiv:1001.2725]. Because exact abundance counts are UV-sensitive (dominated by nearest-neighbor relations), the action used in practice is *smeared* over a small length scale ϵ , which also fixes the overall normalization used throughout this project.

Monte Carlo studies that weight the space of N -element causal sets by $\exp(-\beta * S_{BD}(\epsilon))$ -- a Euclideanized, not oscillatory, weight -- have found a phase transition as β is varied: a continuum-like ("manifold-like") phase at small β whose statistics resemble a genuine Lorentzian sprinkling, and a non-continuum phase at large β dominated by highly-ordered configurations (Kleitman-Rothschild-like orders, or -- in lattice-gas-type models -- literal crystalline layering) [Surya 2012, arXiv:1110.6244]. Locating this transition reliably, and determining its order, is numerically hard: the two phases are separated by a free-energy barrier that single-site Metropolis moves and even naive replica exchange struggle to cross, so a transition can look "located" while actually being an artifact of wherever the sampler's convergence ran out. Section 6 below documents several concrete ways this went wrong during this project before it was caught.

Cunningham & Surya introduce a cheaper alternative to sampling generic N -element orders directly: a *lattice gas*, where N causal-set elements occupy N of the sites of a fixed regular lattice embedded in a d -dimensional cylinder spacetime, and Monte Carlo moves relocate one element to an unoccupied site [Cunningham & Surya 2019, arXiv:1908.11647]. This "dimensionally restricted" construction reproduces the generic model's qualitative manifold-like/layered phase transition at much lower computational cost, but the background lattice is regular -- C&S note explicitly that checking whether this regularity biases the causal structure (as opposed to using "a more realistic model of discreteness", i.e. a randomly sprinkled background) is an open question they do not pursue. Section 5 below is this project's own attempt at that comparison.

2. Models

2D orders with the smeared BD action. A "2D order" on N elements is a pair of independent permutations (u, v) of $\{0, \dots, N-1\}$: element i strictly precedes element j iff $u[i] < u[j]$ AND $v[i] < v[j]$. The intersection of two strict total orders is automatically transitively closed, so the state space is exactly this set of (u, v) pairs, not all labelled partial orders on N elements -- only those realizable as such an intersection (the standard 2D-order representation, e.g. Surya 2012). A Monte Carlo move picks u or v at random and swaps two of its entries (a transposition); at $\beta=0$ this is an unweighted random walk over (u, v) pairs, which samples them uniformly. The smeared 2D action is $S(C, \epsilon) = 4\epsilon * (N - 2\epsilon * \sum_n N_n * f(n, \epsilon))$, $f(n, \epsilon) = (1-\epsilon)^n * (1 - 2\epsilon * n / (1-\epsilon) + \epsilon^2 * n * (n-1) / (2 * (1-\epsilon)^2))$, matched by direct inspection of the fetched paper against this project's implementation, including an $\epsilon \rightarrow 1$

analytic-limit unit test [Glaser, O'Connor & Surya 2018, arXiv:1706.06432; confirmed in `causet_lab/mcmc/action.py` and `results/phase3/mcmc_2d_report.md @ edb9b99`].

Dimensionally-restricted lattice gas. N elements occupy N sites of an $h \times w = m$ regular lattice on a cylinder (metric $ds^2 = -dt^2 + d\theta^2$, $t \in \{0, \dots, h\}$, θ periodic with period w sites), with this project's own $d=2$ sizing convention $w=n$, $h=4n$, $m=4n^2$ matched directly against C&S's own quoted numbers. Moves swap one occupied site with one unoccupied site. **C&S's causal relation is not fully specified in the paper** -- they state 45-degree lightcones and confirm cylinder wraparound exists, but give no integer-coordinate formula; this project derived $(t_a, k_a) \prec (t_b, k_b)$ iff $t_b > t_a$ and $\min(dk, w-dk) \leq t_b - t_a$ (shortest arc on the circumference) and flags this explicitly as its own construction, not a quote [`causet_lab/mcmc/lattice_gas_core.py`, `results/phase4/lattice_gas_2d_report.md @ 0f85d22`]. The smeared action used is C&S's own Eq. 8/9, which has a genuinely different normalization from the order model's action above (prefactor 2ϵ vs 4ϵ , and the interval-count sum starts at $r=1$ vs $r=0$) -- confirmed by direct quotation from both papers, kept as two separate functions rather than unified.

Random-background variant (this project's own construction). Identical region, filling representation, move set, and action; the only change is that the m background sites are a fixed (quenched) uniform-random sprinkling of continuous (t, x) positions instead of sitting on the integer lattice, with x given period w -- the *same* units the lattice's own causal relation uses, not radians (θ has period 2π ; $x = w\theta/(2\pi)$ is the right rescaling to keep the two models' causal geometries identical. See Section 6 for how a units mismatch here was initially missed). **Equivalence check:** placing the background exactly at the regular lattice's own integer positions reproduces `lattice_gas.lattice_to_matrix` bit-for-bit, including the lightlike ($\leq$) convention, across 3 sizes and 5 seeds each [`causet_lab/tests/test_random_bg.py::test_random_bg_at_lattice_positions_matches_lattice_exactly @ 1ff7ef3`; all 3 parametrizations pass].

3. Methods

Incremental bitset updates. Each element's relation to every other is tracked as a `uint64` bitset (`future[x] / past[x]`); interval size for a related pair is `popcount(future[x] & past[y])`. A move changes only the relations of the element(s) that moved, updated via $O(N)$ scalar bookkeeping rather than an $O(N^2)$ matrix rebuild -- validated bit-for-bit against full recomputation over $1e5$ random moves for every model in this project (orders, lattice gas, random background) [`causet_lab/tests/test_bitset_core.py`, `test_lattice_gas.py`, `test_random_bg.py`]. This caps all three models at $N \leq 64$ (one element's relations fit in one machine word); see Section 7.

Parallel tempering (PT). The original (Phase 3) approach: replicas at a ladder of beta values, with swap moves between neighbors and round-trip tracking to confirm replicas actually cross the transition rather than merely reporting a locally-plausible swap rate [`causet_lab/mcmc/tempering.py`]. The method traces to exchange Monte Carlo / replica exchange [Swendsen & Wang, Phys. Rev. Lett. 57, 2607 (1986)]; this project implemented it from the general replica-exchange description rather than fetching and checking a specific paper's equations against the code (unlike the citations marked `@ commit` elsewhere in this paper), so it is cited by name only, without that same code-level verification.

Wang-Landau + multicanonical (WL/MUCA). Replacing PT (see Section 6 for why): Wang-Landau [Wang & Landau, Phys. Rev. Lett. 86, 2050 (2001)] estimates the action density of states $g(S)$ by an adaptive-weight random walk that flattens the visited-bin histogram across the whole reachable range (stage-doubling

modification factor, with stall detection distinguishing a genuinely unreachable bin from a merely slow one); multicanonical production then samples with that weight frozen and reweights the single resulting run to the canonical $P_{\beta}(S)$ and β at *any* β , using the Berg-Neuhaus histogram-corrected estimator $P(S) \sim H_{\text{muca}}(S) * \exp(\ln_g(S) - \beta * S)$ [Berg & Neuhaus, "Multicanonical algorithms for first order phase transitions", Phys. Lett. B 267 (1991) 249 -- no arXiv ID on record for this pre-arXiv-era reference]. β_c is located as the action-variance (specific-heat) maximum, matching the definition Glaser-O'Connor-Surya themselves use, not an equal-peak-height double-peak criterion (kept as a separate diagnostic) [Glaser, O'Connor & Surya 2018, Sec. 4.1].

Windowed round-trip counting. A round trip is counted between the two *physically relevant* reachable extremes, which can be narrower than the full `ever_visited` span when most of $P_{\beta}(S)$'s mass for the β range of interest sits well inside it (Section 6).

Checkpointing. Atomic pickle writes of the full walker state (configuration, splitmix64 RNG state, WL/MUCA weights and histograms), exercised in practice by intentionally interrupting and resuming runs [`causet_lab/mcmc/checkpoint.py`].

Uncertainty. Two independent estimators are used depending on what a given run produced: a block-bootstrap over round-trip-sized windows of a single run's time series (used for the $N=30$, $\text{eps}=0.5$ orders result), or the empirical spread across independent parallel workers' own located β_c (used for $N=40$, $\text{eps}=0.5$ and, in Section 5, across background realizations).

4. Validation

2D orders, $\text{eps}=0.5$, $N=30$ and $N=40$: β_c vs. the published formula. The formula is $\beta_c(N, \text{eps}) = b(\text{eps})/N + c(\text{eps})/N^2$, $b(\text{eps})=1.66(+/-0.03)/\text{eps}^2$, $c(\text{eps}) = 4.09(+/-0.50)/\text{eps}^3 - 27.77(+/-2.45)/\text{eps}^2$ [Glaser, O'Connor & Surya 2018]. The leading term alone is a poor approximation at these sizes -- the $1/N^2$ correction is not a small perturbation here, consistent with the paper's own statement that its asymptotic regime is only reached for $N \gg 65$:

N	leading term b/N	full formula	measured β_c	z
30	0.2213	0.1343 +/- 0.0158	0.1478 +/- 0.0004 (bootstrap)	0.86
40	0.1660	0.1170 +/- 0.0096	0.1221 +/- 0.0011 (4-worker spread)	0.52

[`results/phase3b/muca_calibration_report.md @ 7d648fa` ; leading/full formula values recomputed directly from the published coefficients above for this table]. Both PASS ($z < 3$), though see Section 7 for why this should not be read as confirming the paper's full first-order picture.

2D orders, $\text{eps}=0.1$, $N=30$: independent cross-check. Run fresh (not part of the original $\text{eps}=0.5$ calibration) as a side check during the lattice-gas work below: located $\beta_c = 6.939$, full-formula prediction $6.992 +/- 0.627$ (leading term alone: 5.533) -- within about 0.1 formula-standard-deviations, with no bootstrap on the measured side for this single run [`results/phase4/lattice_gas_2d_report.md @ 0f85d22` , Part 5].

PT vs. MUCA at $N=30$, $\text{eps}=0.5$: not independent confirmation. The original PT scan located $\beta_c = 0.14797 +/- 0.00384$ at this (N, eps) [`results/phase3/mcmc_2d_report.md @ edb9b99`], close to the WL/MUCA value above ($0.1478 +/- 0.0004$, agreement to within 0.001

[results/phase3b/muca_calibration_report.md @ 7d648fa]) -- but the PT value is flagged in its own source report as right-censored (the largest converged beta in that scan, not a confirmed interior maximum; see Section 6), so it cannot serve as an independent check on the WL/MUCA result. This project's actual validation of beta_c rests on the WL/MUCA-vs-published-formula comparison in the table above, not on PT/MUCA agreement.

Lattice gas 2D: controls and qualitative phase comparison. beta=0 (uniformly random fillings) sampled as sub-intervals (since the cylinder filling is not itself an Alexandrov interval) give Myrheim-Meyer dimension 2.292 (n=30) / 2.073 (n=50), both close to the d=2 target, and interval-abundance L1 distance to a true sprinkle of 0.100 / 0.086 [results/phase4/lattice_gas_2d_report.md @ 0f85d22]. At n=30, the located beta_c = 11.118 separates a hot phase (ordering fraction 0.882, matching C&S's own quoted hot-phase value of 0.88 almost exactly; MM dimension 2.18; abundance-profile distance to a sprinkle 0.065) from a cold phase at 4*beta_c (ordering fraction 0.646; MM dimension 2.87; abundance-profile distance 0.794) -- every observable moves in C&S's reported direction, though the cold-phase ordering fraction does not reach their asymptotic ~0.5 bilayer value at this much smaller n [same source, Part 4].

P_beta_c(S), described cautiously. The pooled action histogram at the located beta_c is a single broad hump at both N=30 and N=40 (eps=0.5), not a resolved double peak [results/phase3b/muca_calibration_report.md @ 7d648fa , plot muca_P_beta_c_S_eps0p50.png]. This is consistent with either a genuinely weak/crossover-like transition at these small N (both are well below the paper's own $N \gg 65$ asymptotic regime) or with insufficient statistics to resolve a real but small barrier -- **the data collected here cannot distinguish the two**, and this should not be read as a confirmed first-order signature at either N.

5. New result: random vs. regular background

Stage 1: n=30, 5 independent background realizations (sprinkled once per realization, then held fixed -- "quenched" -- for that realization's entire WL+MUCA run), run in parallel across 4 CPU cores, total wall time 1623.9s. The background seed and the Metropolis-chain seed are always independent parameters for a given realization (never derived from one another), so the only thing varying across the 5 runs is the background geometry itself. Same eps=0.1, same $m = 4*n^2$, same WL/MUCA/checkpointing pipeline as the Phase 4 regular-lattice run, with the beta_c-location scan bracket deliberately widened (0.01x to 30x the regular lattice's own located beta_c, used only as a pilot-range anchor, not an assumption the two models share a beta_c) since there is no a priori reason quenched disorder should leave beta_c unchanged [results/phase5/random_background_report.md @ 3ae1ad5 , this section].

Equivalence check: placing the background exactly at the regular lattice's integer positions reproduces lattice_to_matrix bit-for-bit across 3 sizes and 5 seeds each -- confirms the random background's causal geometry is identical to the regular lattice's at those positions, not merely "the same region" (see Section 6 for the units bug this caught).

Control: beta=0 random-background fillings (5 seeds, same sub-interval-sampling methodology as Phase 4) give MM dimension 2.267 (n=30) / 2.061 (n=50), abundance-profile distance to a sprinkle 0.089 / 0.060, and whole-filling ordering fraction 0.8768 / 0.8766 -- matching the regular lattice's own n=30 beta=0 values (MM dim 2.292, distance 0.100, ordering fraction ~0.882) at least as well.

beta_c: tightly reproducible across realizations (9.709, 9.757, 9.756, 9.766, 9.627; mean 9.723, std 0.058, SEM 0.026). Checked against a matched-pipeline comparison, not a single-seed one: the regular lattice re-run through the identical `random_bg` code path (positions fixed at the integer grid, otherwise the exact same settings) at 5 independent chain seeds gives 11.095 +/- 0.005 (SEM, std 0.011) -- consistent with Phase 4's original single-seed value (11.118) to within about 2 std. Gap = 1.372, combined SEM = 0.0265, **z = 51.8: significantly below** the regular lattice, not explainable by single-seed noise on either side. An unexpected side finding from the matched-pipeline rerun: the regular lattice mixed markedly worse under Wang-Landau than every random-background realization (total wall time 6659.9s for 5 chain seeds vs. 1623.9s for 5 backgrounds) -- plausibly the lattice's exact periodic structure supports slow modes that quenched disorder breaks up, not independently confirmed.

Mechanism check: does the random background simply reach a deeper cold-phase action floor (which would mechanically lower beta_c)? Checked directly against both backgrounds' actual WL/MUCA data: lowest action reached (-2.4716 +/- 0.0013 regular vs. -2.4723 +/- 0.0008 random) and hot-phase mean action (4.3632 +/- 0.0004 vs. 4.3597 +/- 0.0081) are statistically indistinguishable between the two. **The hypothesis does not hold** -- the beta_c shift is not explained by the random background being able to get more deeply ordered; the real mechanism (presumably a difference in the density-of-states shape across the explored range, not its extremes) is left open.

Hot- vs. cold-phase observables (hot: 5 fresh beta=0 fillings per background; cold: 5 independent chains per background annealed to 4 * that background's own beta_c ; values are mean +/- std across the 5 backgrounds):

Observable	Hot	Cold	Reg. lattice, hot	Reg. lattice, cold
MM dimension	2.163 +/- 0.145	2.729 +/- 1.041	2.18	2.87
Ordering fraction	0.876 +/- 0.007	0.583 +/- 0.022	0.882	0.646
Height	7.00 +/- 0.51	3.44 +/- 0.36	7.40	4.20
Occupied rows (/120)	26.32 +/- 0.41	13.16 +/- 0.61	26.7	12.8

The hot phase is close to background-independent (every value within about 1 std of the regular lattice's). The cold phase differs measurably: lower ordering fraction and height on the random background than the regular lattice at a comparably-deep point relative to each model's own beta_c (not a matched absolute beta, by construction) -- in the direction of C&S's own asymptotic bilayer limit (~0.5), slightly closer here than the regular lattice got at n=30.

P_beta_c(S) shape: a naive check using this project's existing peak-finder flagged a "double peak" with a large barrier (17-21) at 4/5 realizations -- **confirmed spurious** on inspection (one candidate "peak" in every case was 8-11 orders of magnitude below the real mode, see Section 6), and **fixed upstream** in `muca._peak_diagnostics` itself (a new mass-fraction filter, with regression tests) rather than left as a local workaround. After the fix, all 5 realizations show a single broad hump, the same picture as the regular lattice and the orders model. Re-checking Phase 3b's and Phase 4's committed conclusions against their actual saved checkpoints confirms the fix changes neither (both still single-hump) -- though it does retroactively show that Phase 4's lattice-gas P_beta_c(S) would have triggered the *same* spurious double peak had anyone checked it at the time; nobody had. **No confirmed first-order signature at any realization or model examined in this project.**

Plain answer (n=30): the transition's *location* depends on the background being a regular grid (a real ~12% shift, confirmed against a matched 5-seed regular-lattice ensemble, not a single-seed artifact, and not explained by the simplest mechanism checked, well outside the cross-realization spread); the transition's *qualitative character* does not (manifold-like/ordered phases persist on both, the hot phase is quantitatively close to background-independent, neither background shows a resolved double peak). Reported as found, not rounded toward either "no effect" or "a different transition."

Stage 2: n=50. Getting a trustworthy result here took longer than expected -- two further sampler bugs, both general lessons for this WL pipeline, are reported in full in Section 6. In brief: (1) the histogram-widening step filled new bins with a flat repeated value instead of a gradient, which could open an unphysical `ln_g` cliff at the far edge of a widened block under ordinary differential visitation (found directly: three of the first ten n=50 runs produced adjacent-bin `ln_g` steps of 101-66,767, vs. 6-11 for every healthy run, each leaving its MUCA production stuck at zero round trips after 300,000,000 moves); (2) after fixing that, one seed hit the *same* failure signature again, from a genuinely kinetically-isolated bin -- entropically ordinary, but reachable by almost no accepted single-site move from the bulk -- which no amount of smarter `ln_g` seeding can fix, since the problem is reachability, not weight. Fixed by replacing adaptive widening with a *fixed* sampling window set from pilot quantiles (hot end: 99.99th percentile of a beta=0 sample; cold end: 0.01st percentile of a sample annealed to 3.5x the beta_c anchor), verified post-hoc to carry negligible reweighted probability mass (<1e-3, typically <1e-4) at either edge for beta in [0, 2*beta_c] on every production run.

With that fixed (and a direct two-seed check confirming the methodology change does not alter beta_c for seeds that were already clean -- differences of 0.002 and 0.008, well inside each ensemble's own seed spread), n=50 gives beta_c = 5.959 +/- 0.033 (SEM) for the random background vs. 6.419 +/- 0.005 (SEM) for the regular lattice. **Gap = 0.460, z = 13.67** -- still overwhelmingly significant, but down sharply from n=30's z=51.8. The drop in z is driven mainly by the *absolute* gap itself shrinking about 3x (1.372 at n=30 -> 0.460 at n=50): holding the n=30 combined SEM fixed, that gap alone would give z ~ 17.4, most of the way from 51.8 to the actual 13.67. The random background's larger cross-seed spread at n=50 (SEM 0.033 vs. 0.026) is a secondary contributor, not the main driver. In relative terms the shift is **7.2%, down from n=30's 12.4%** -- and that relative shift times n is nearly constant between the two sizes (12.4%x30=371 vs. 7.2%x50=358, a 3.6% difference), consistent with the gap itself falling off roughly as 1/n over this two-point baseline (two-point power-law exponent ~1.07). The cold-phase ordering-fraction gap that was clearly present at n=30 (0.583 vs. 0.646) has nearly closed at n=50 (0.661 vs. 0.660); the cold-phase MM dimension converges too (2.73/2.87 -> 3.60/3.60). The hot phase remains close to background-independent, and all 10 n=50 P_beta_c(S) curves (5 per background) are single broad humps, no double-peak signature at either background [results/phase5/random_background_report.md @ 743a94b , Stage 2 section].

Read plainly: the background-dependence of the transition looks like a finite-size effect that fades with system size, not a permanent discrepancy between the two models -- though two data points (n=30, n=50) cannot by themselves distinguish that from a slower-than-1/n falloff that plateaus above zero. Taking the 1/n reading at face value and anchoring at n=50 predicts a relative gap of about **5.6% at n=64** (7.2% x 50/64) -- stated here as a concrete, checkable number for whoever runs that size next, currently blocked on the N<=64 bitset limit (Section 7/8).

6. Pitfalls and lessons

Right-censored "peaks" and the blind grid-recentering test. The original PT scan's `beta_c` was, for 7 of 8 tested (N, eps) combinations, exactly the largest *converged* beta in its scan -- i.e. there was no converged data confirming the variance actually turns back down past the located point. The original search grid was itself only centered at about 0.42x of the published formula's own prediction, not at the prediction itself. A direct stress test (N=40, deliberately re-centering the grid independently at 0.6x and 1.6x of that same formula prediction -- not relative to the original 0.42x point) found that at eps=0.21 the located `beta_c` relocated to ~7.0 from both of those re-centered grids -- nowhere near the original grid's own 0.70 or the published 0.82 -- while eps=0.5 held up (all three grid placements landed within 25% of the prediction) [`results/phase3/mcmc_2d_report.md @ edb9b99`]. Lesson: a `beta_c` that sits at the edge of whatever range was searched is not located; it is censored, and should be reported as such until an interior maximum is confirmed, grid placement included.

The `beta_c` definition mismatch. An early version of the first-order diagnostic used equal-peak-height, not the action-variance maximum that Glaser-O'Connor-Surya's own formula was fit to -- checked directly against their Sec. 4.1 and corrected before any formula comparison was trusted [`causet_lab/mcmc/muca.py @ 7d648fa`].

Noise mistaken for a double peak. A naive peak finder accepted any two local maxima regardless of the dip between them, producing a false "double peak" at N=30 with a barrier of only 0.004-0.07 across an 800-point beta scan -- a <7% dip on a single broad hump, not a first-order signature. Fixed by requiring a minimum barrier (`MIN_BARRIER_FOR_BIMODAL = 0.5`) before accepting a two-peak structure [same source].

A reweighting estimator blind to its own weight error. The original `reweight_P_beta` assumed `ln_g` already equalled the true log density of states -- true only once Wang-Landau has fully converged. Fixed by folding in the actual MUCA production histogram (`reweight_P_beta_corrected` , the standard Berg-Neuhaus correction), which stays exact even with an imperfect weight provided the walk visits the region being compared [same source].

Full-span vs. windowed round trips. At N=40, requiring a round trip to reach the full `ever_visited` span (up to S=+402) undercounted real excursions, since essentially all of `P_beta(S)`'s mass for the beta range being tested sat in a much narrower window (bins 0-85 of 126). Recounting the same run with that window instead of the full span found 4x more round trips in under half the wall-clock time of a fresh run -- same physics, same sampling, only the counting criterion changed [same source].

The units bug in the first random-background draft. The initial implementation sprinkled the spatial coordinate as `theta` in radians (period 2π) and compared it directly against `dt` in the lattice's own integer row units -- silently rescaling the lightcone by a factor of $w/(2\pi)$ relative to the regular lattice's own causal relation, which compares an integer step count (period `w`) against `dt` directly. Caught before any sampling was run, by an explicit request to check units and add a same-positions equivalence test against the regular lattice (Section 2); fixed by sprinkling a coordinate `x` with period `w` instead. Lesson: "the same region" is not automatically "the same causal geometry" once a discrete model is replaced by a continuous one -- an explicit reduces-to-the-known-case test is the check that would have caught this immediately, and should be written *before* any production sampling, not after.

A second, different way to mistake noise for a double peak. The existing `MIN_BARRIER_FOR_BIMODAL` check (above) guards against a *shallow* dip on one broad hump. It does not guard against a *deep* dip where one side of the "peak pair" is simply numerical noise in an exponentially suppressed tail: in the random-background analysis (Section 5), 4 of 5 realizations' `P_beta_c(S)` were flagged as double-peaked with a large barrier (17-21), but the "second peak" in every case sat 8-11 orders of magnitude below the real mode --

`_peak_diagnostics`'s `peak_val = 0.5*(p[i1]+p[i2])` average is dominated by the real peak regardless of how negligible the other one is, so the barrier calculation never actually tests whether both candidate peaks carry physically meaningful probability mass. **Fixed upstream** this time, not left as a local workaround: a mass-fraction filter (`MIN_PEAK_MASS_FRACTION`, each candidate peak must exceed 1% of the mode) was added directly to `_peak_diagnostics` [`causet_lab/mcmc/muca.py`], with three regression tests covering the deep-tail-bump rejection, a genuine two-peak case, and the boundary [`causet_lab/tests/test_mcmc.py`]. Re-run against Phase 3b's and Phase 4's own saved checkpoints: both still report no double peak (unchanged); Phase 4's lattice-gas `P_beta_c(S)` would have shown the *same* spurious double peak (barrier=22.3) had anyone run this check on it at the time -- nobody had, so there was no committed claim to correct, but it confirms the bug was general, not specific to disorder.

A histogram-widening step that filled new bins flat instead of with a gradient. When Wang-Landau's bin range needs to grow (edge-hit pressure), the new block of bins was seeded with a single repeated value (the old edge bin's `ln_g`) rather than a gradient continuing the locally observed trend. The bin immediately adjacent to the old edge matched its neighbor fine, but bins further into that artificially flat block could drift apart from each other under ordinary differential visitation; once the walker reached the *far* edge of that drift, a large, unphysical `ln_g` cliff could open there. Found directly at `n=50`: three of the first ten runs produced adjacent-bin `ln_g` steps of 101-66,767 (vs. 6-11 for every healthy run), each exactly at a widened block's far edge, each leaving MUCA production stuck at zero round trips after the full 300-million-move cap. Fixed by extrapolating the locally-observed slope instead of repeating the edge value, clamped so the extrapolation can never exceed the edge value (direction) or exceed the single largest step actually observed nearby (magnitude) [`causet_lab/mcmc/muca.py`, `_widen`, with regression tests in `test_mcmc.py`].

A kinetically-isolated bin that no amount of reweighting can fix. After the fix above, the same failure signature recurred on one seed -- a bin reached only well into the run, not created by any widening event. Direct inspection (resuming the exact frozen walker state and tabulating where proposed moves actually land) found the bin's nearby entropy was unremarkable, but almost no single-site relocation from the bulk landed there with any acceptable probability: it was *kinetically*, not thermodynamically, hard to reach. Wang-Landau's histogram-flattening heuristic reads sustained under-visitation as "needs more weight," which only helps when the cause is low entropy -- it cannot fix a reachability problem, since the walker still has to get there via an accepted move regardless of that bin's weight. Fixed architecturally, not by better seeding: the sampling window is now fixed from pilot quantiles rather than left free to adapt, so the walker is kept away from the pathological region entirely, with a post-hoc check (`verify_window_coverage_randombg`) confirming nothing physically relevant was cut off [`causet_lab/mcmc/random_bg.py`].

7. Limitations

- **Small sizes.** `N`, `n` ≤ 60 throughout, but not uniformly: the legacy PT beta-ladder scan (Section 6) reached `N=30-60`; the WL/MUCA results that replaced it (Section 4) only cover `N=30` and `N=40` for the orders model, and `n=30` and `n=50` for the lattice gas and random background (Section 5) -- no WL/MUCA run went past `N=40` / `n=50`. The bitset engine's `uint64` -per-element representation caps every model in this project at `N`, `n` ≤ 64 without a redesign (see Outlook).
- **No cleanly resolved first-order barrier at any size tested.** Hysteresis disagreed between random- and layered-start chains for 6/8 (`N`, `eps`) PT combinations, the double-peak histogram signature appeared in only 1/8, and MUCA's own `P_beta_c(S)` is a single broad hump at both `N` tested

(Section 4). Whether this reflects genuinely weak/crossover behavior below the paper's asymptotic regime, or insufficient statistics, is not resolved here.

- **Single epsilon for most results.** $\text{eps}=0.5$ is the primary orders- model coupling; $\text{eps}=0.1$ and $\text{eps}=0.21$ appear only as side checks or ($\text{eps}=0.21$) in the earlier, since-superseded PT study. The lattice-gas and random-background models use only $\text{eps}=0.1$, matching C&S's own choice. Section 4's full-formula vs. leading-term comparison is only tabulated at $N=30$ for $\text{eps}=0.1$ (not $N=40$); the $\text{eps}=0.5$ table has both.
- **Few seeds.** Most orders-model and lattice-gas results are single- seed; Phase 5 (Section 5) is this project's first result with an explicit multi-seed spread, now at two sizes ($n=30$, $n=50$). The $n=50$ dataset mixes two sampler methods by seed (5 of 10 re-run under a fixed-window method after two bugs were found, 5 kept from the original run) -- justified by a direct two-seed consistency check (0.002 and 0.008 differences, well inside each ensemble's own spread), not by assumption, but a fully single-method $n=50$ dataset was not run. The $n=50$ moves-per-stage/moves-per-round-trip mixing comparison (Section 5/Appendix) rests on only 3 of 5 random- background seeds and 2 of 5 regular-lattice seeds, since the other seeds' per-stage logs were not retained -- too few for a confident per-move mixing-speed claim between backgrounds at $n=50$, reported as inconclusive rather than rounded to a conclusion.
- **Only two system sizes for the background-dependence finding.** The observed gap shrinks from 12.4% ($n=30$) to 7.2% ($n=50$), consistent with a finite-size effect, but two points cannot distinguish that from a slower falloff that plateaus above zero. The $n=64$ test in Section 8 would help distinguish these.
- **Two open mechanism questions, neither resolved by the $n=50$ work.** What actually causes the β_c shift (Section 5's mechanism check only rules out the simplest explanation, a deeper cold-phase action floor, without identifying the real cause) and why the regular lattice mixed 4x slower than the random background at $n=30$ under Wang-Landau (Section 5) both remain open; the $n=50$ wall-clock data is confounded by the sampler-method mix above and was not used to re-test the 4x finding either way.
- **Euclideanized weights throughout.** Every model here samples $\exp(-\beta S_{\text{BD}})$ as a real, non-oscillatory Boltzmann weight -- a standard device in this literature, not a Lorentzian quantum path integral with genuine phases. Nothing in this project addresses that gap.

8. Outlook

- **Multiword bitsets.** Extending each element's relation bitset from one `uint64` to two or more machine words would lift the $N \leq 64$ cap that currently excludes $n=80$ for the lattice gas (explicitly out of scope in Phase 4) and any $N > 64$ orders-model run.
- **Parallel / windowed Wang-Landau.** Phase 3b's $N=40$ multicanonical recursion plateaued at 1-2 round trips per 30M moves across 8 iterations without converging; a windowed, multi-walker WL scheme (rather than single-chain recursion) is the natural next attempt, informed by the round-trip-window fix in Section 6.
- **3D reproduction.** Nothing in this project yet repeats the 2D orders calibration in $d=3$; this would test whether the same right-censoring/blind-regridding failure modes recur in a higher- dimensional order space.
- **4D lattice gas.** C&S's own paper covers $d>2$ lattice gases; this project has only built $d=2$.
- **A cost estimate from a scaling ladder.** Phase 3b found mixing cost rising steeply from $N=30$ to $N=40$ (10 round trips in 508s of production vs. 12 round trips needing a windowed-counting fix after a

~1750s WL phase and a non-converging ~3850s recursion phase); a deliberate $N=30/40/50/60$ (or $n=30/50/80$, once multiword bitsets exist) ladder, timed consistently, would let a real extrapolated cost curve replace this project's current ad hoc per-size reporting. Phase 5's $n=30/n=50$ pair is a start for the random-background model specifically, but used two different sampler methods by seed (Section 6/7), so it is not yet the clean, consistently-timed ladder this item calls for.

- **A third system size for the background-dependence gap: the $n=64$ test.** The observed shrinkage (12.4% $\rightarrow$ 7.2%) is consistent with a finite-size effect but is only two points. $n=64$ is reachable with no code changes (it is exactly the current $N_{\leq 64}$ bitset ceiling) and gives a concrete, falsifiable prediction to check: a simple $1/n$ falloff anchored at $n=50$ predicts a relative gap of about **5.6%** (7.2% $\times$ $50/64$); anything well outside that -- a gap that has plateaued above zero, or one that has already closed -- would rule out the simple $1/n$ reading. $n=80$ would help further but needs the multiword-bitset extension above first.

Code availability

All code, tests, scripts, result data, and reports cited in this paper live in a single repository (see `README.md` at the repository root for the folder map, installation, and reproduction instructions for every number and figure above). A public URL had not been assigned as of this writing; the author is responsible for adding it here once the repository is published -- see the project README's by-hand checklist.

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Appendix: run parameters, timings, hardware

Hardware: single desktop (HP Pavilion Desktop 590-p0xxx), Intel Core i3-8100 @ 3.60GHz, 4 cores / 4 logical processors (no hyperthreading), Windows 11 Home (build 10.0.26200). Python 3.13.2, NumPy 2.3.2, Numba 0.68.0.

Per-run timings (single seed unless noted; WL = Wang-Landau phase, MUCA = multicanonical production phase, target 10 round trips throughout):

Model	N/n	eps	WL	MUCA	Total	Source
2D orders (PT, legacy)	30-60	0.21, 0.5	--	--	not separately timed (beta-ladder scan)	<code>results/phase3/mcmc_2d_report.md @ edb9b99</code>

Model	N / n	eps	WL	MUCA	Total	Source
2D orders (WL/MUCA)	30	0.5	~1700s	508s	~37 min	results/phase3b/muca_calibration_report.md @ 7d648fa
2D orders (WL/MUCA)	40	0.5	~1750s	457.3s (windowed)	--	same
2D orders (WL/MUCA)	30	0.1	735.5s	150.1s	885.6s	results/phase4/lattice_gas_2d_report.md @ 0f85d22
Lattice gas	30	0.1	1613.7s	948.2s	2561.8s (~42.7 min)	same
Random background (5 realizations)	30	0.1	387.3-1435.5s	141.1-275.6s	1623.9s total (all 5, 4 cores)	results/phase5/random_background_report.md @ 3ae1ad5
Regular lattice, matched pipeline (5 chain seeds)	30	0.1	1345.0-5704.3s	511.1-1057.9s	6659.9s total (all 5, 4 cores)	same report, matched-pipeline rerun section
Random background (5 realizations, mixed method)	50	0.1	1180.3-2745.1s	245.6-4178.4s	17590.5s total (all 5, 4 cores)	results/phase5/random_background_report.md @ 743a94b , Stage 2 section
Regular lattice via rbg pipeline (5 chain seeds, mixed method)	50	0.1	1291.7-2765.9s	281.5-1398.7s	12194.7s total (all 5, 4 cores)	same, Stage 2 section

Multiprocessing: Phase 5 stage 1 runs 5 independent realizations via

`ProcessPoolExecutor(max_workers=4)` -- 4 concurrent, the 5th queued behind whichever finishes first. The matched-pipeline regular- lattice rerun took **4x longer in total** than the random-background ensemble under identical settings -- see Section 5's mechanism-check discussion and Section 6 (not fully explained; flagged as an open observation, not a settled finding). The n=50 totals are **not** a clean replay of that comparison: each model's 5-seed total mixes two sampler methods (some seeds re-run under a fixed-window method after two bugs were found, the rest kept from the original adaptive-widening run -- the exact split differs slightly by model; see Section 6/7), so the n=50 row is not used to re-test the n=30 mixing asymmetry.

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